

7. The chemical industry provides many of the chemicals that people need for modern life. The chemical industry today is developing new processes to manufacture these chemicals more efficiently and with less impact on the environment.

(a) Explain why catalysts allow chemical reactions to be carried out at lower temperatures than would otherwise be needed. [2]

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(b) Describe the economic and environmental benefits of developing new and better catalysts. [3]

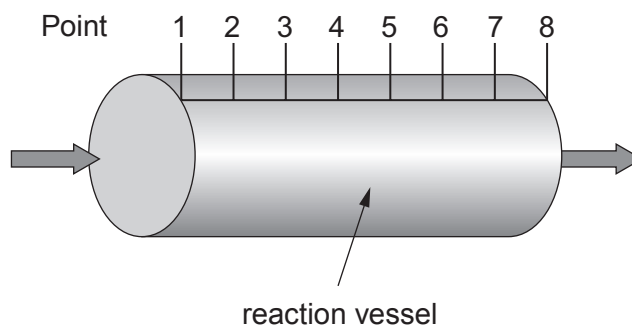
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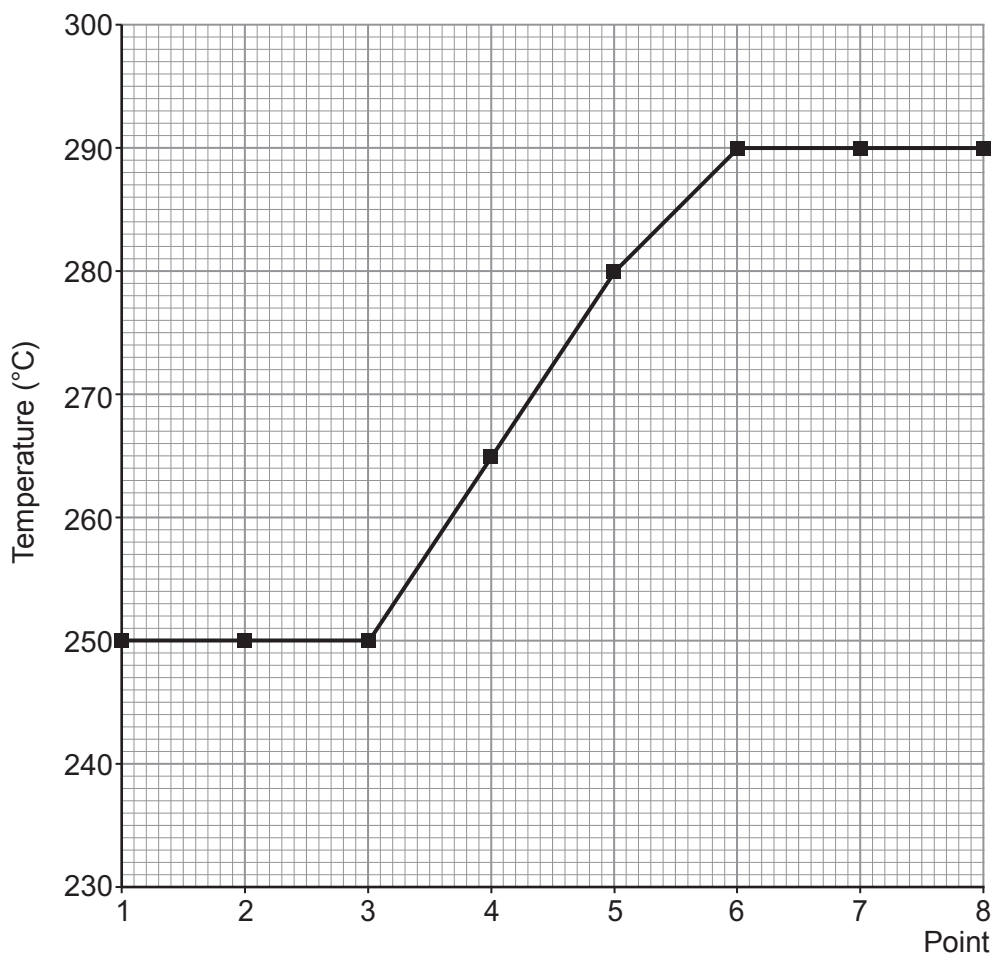
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(c) Analysis of a reaction gives a good measurement of catalyst performance.



In an industrial process, waste gases, which include hydrogen sulfide, are pumped through a reaction vessel containing a catalyst. The hydrogen sulfide is removed in an **exothermic** reaction which causes an increase in gas temperature. The catalyst is performing badly if the hydrogen sulfide is not removed. Thermocouples monitor the temperature across a reaction vessel at eight different points as shown above.

The temperature at each of the points 1 to 8 in the reaction vessel are plotted on the graph below.



Evaluate the performance of the catalyst across the reaction vessel.

[3]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1														4 He Helium 2			
														19 F Fluorine 9			
														16 O Oxygen 8			
														14 N Nitrogen 7			
														12 C Carbon 6			
														11 B Boron 5			

1 H Hydrogen 1

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number